17

Sound Waves

CHAPTER OUTLINE

17.1 Pressure Variations in Sound Waves

17.2 Speed of Sound Waves

17.3 Intensity of Periodic Sound Waves

17.4 The Doppler Effect

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ17.1 Answer (b). The typically higher density would by itself make the speed of sound lower in a solid compared to a gas.

OQ17.2 Answer (e). The speed of sound in air, at atmospheric pressure, is determined by the temperature of the air and does not depend on the frequency of the sound. Sound from siren A will have a wavelength that is half the wavelength of the sound from B, but the speed of the sound (the product of frequency times wavelength) will be the same for the two sirens.

OQ17.3 Answer (c). The ambulance driver, sitting at a fixed distance from the siren, hears the actual frequency emitted by the siren. However, the distance between you and the siren is decreasing, so you will detect a frequency higher than the actual 500 Hz.

OQ17.4 Answer (d). When a sound wave travels from air into water, several properties will change. The wave speed will increase as the wave crosses the boundary into the water causing the spacing between crests (the wavelength) to increase, because crests move away from the boundary faster than they move up to the boundary. The sound intensity in the water will be less than it was in air because some sound is reflected by the water surface. However, the frequency (number of crests passing each second) will be unchanged, since a

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crest moves away from the boundary every time a crest arrives at the boundary.

OQ17.5 Answer (d). The drop in intensity is what we should expect according to the inverse-square law:

$$\frac{I_1}{I_2} = \frac{r_1^2}{r_2^2}: \quad \frac{2\ \mu\text{W/m}^2}{0.2\ \mu\text{W/m}^2} = 10 = \frac{(950\ \text{m})^2}{(300\ \text{m})^2}$$

OQ17.6 Answer (d). We have f_s = 1 000 Hz, v = 343 m/s, v_o = −30 m/s, v_s = 50 m/s. We find

$$f' = \frac{f(v + v_o)}{(v - v_s)} = \frac{(1\ 000\ \text{Hz})[(343\ \text{m/s}) + (-30\ \text{m/s})]}{343\ \text{m/s} - 50\ \text{m/s}}$$
$$= 1\ 068\ \text{Hz}$$

OQ17.7 Answer (b). A sound wave is a longitudinal vibration that is propagated through a material medium.

OQ17.8 (i) Answer (b). The frequency increases by a factor of 2 because the wave speed, which is dependent only on the medium through which the wave travels, remains constant.

(ii) Answer (c).

OQ17.9 Answer (a) We suppose that a point source has no structure, and radiates sound equally in all directions (isotropically). The sound wavefronts are expanding spheres, so the area over which the sound energy spreads increases according to $A = 4\pi r^2$. Thus, if the distance is tripled, the area increases by a factor of nine, and the new intensity will be one-ninth of the old intensity. This answer according to the inverse-square law applies if the medium is uniform and unbounded. For contrast, suppose that the sound is confined to move in a horizontal layer. (Thermal stratification in an ocean can have this effect on sonar "pings.") Then the area over which the sound energy is dispersed will only increase according to the circumference of an expanding circle: $A = 2\pi rh,$ and so three times the distance will result in one-third the intensity. In the case of an entirely enclosed speaking tube (such as a ship's telephone), the area perpendicular to the energy flow stays the same, and increasing the distance will not change the intensity appreciably.

OQ17.10 (i) Answer (c). Both observer and source have equal speeds in opposite directions relative to the medium, so in $f' = (v + v_o)/(v - v_s)$ we would have something like $(343 - 25)f/(343 - 25) = f$.

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(ii) Answer (a). The speed of the medium adds to the speed of sound as far as the observer is concerned, to cause an increase in $\lambda = v/f$. The wind "stretches" the wavelength out.

(iii) Answer (a).

OQ17.11 In order of decreasing size we have (b) > (d) > (a) > (c) > (e). In $f' = f\,[(v+v_o)]/[(v-v_s)]$ we can consider the size of the fraction $(v+v_o)/(v-v_s)$ in each case, where the positive direction for the observer is toward the source, the positive direction for the source is toward the observer: (a) 343/343 = 1, (b) 343/(343 − 25) = 1.08, (c) 343/(343 + 25) = 0.932, (d) (343 + 25)/343 = 1.07, (e) (343 − 25)/343 = 0.927.

OQ17.12 Answer (c). The intensity is about $10^{-13}\ \mathrm{W/m^2}$.

OQ17.13 Answer (c). Doubling the power output of the source will double the intensity of the sound at the observer's location. The original decibel level of the sound is $\beta = 10 \cdot \log(I/I_0)$. After doubling the power output and intensity, the new decibel level will be

$$\begin{aligned}\beta' &= 10 \cdot \log(2I/I_0) = 10 \cdot \log\left[2(I/I_0)\right] = 10 \cdot \left[\log(2) + \log(I/I_0)\right] \\ &= 10 \cdot \log(2) + \beta\end{aligned}$$

so the increase in decibel level is $\beta' - \beta = 10 \cdot \log(2) = 3.0\ \mathrm{dB},$ making (c) the correct answer.

OQ17.14 Answer (c). The threshold of human hearing is defined as 0 dB; the average person cannot hear sound with a lower intensity level. Normal conversation has an intensity level of about 60 dB.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ17.1 For the sound from a source not to shift in frequency, the radial velocity of the source relative to the observer must be zero; that is, the source must not be moving toward or away from the observer. The source can be moving in a plane perpendicular to the line between it and the observer. Other possibilities: The source and observer might both have zero velocity. They might have equal velocities relative to the medium. The source might be moving around the observer on a sphere of constant radius. Even if the source speeds up on the sphere, slows down, or stops, the frequency heard will be equal to the frequency emitted by the source.

CQ17.2 The speed of sound in air is proportional to the square-root of the absolute temperature, $\sqrt{T}$. The speed of sound is greater in warmer

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air, so the pulse from the camera would return sooner than it would on a cooler day from an object at the same distance. The camera would interpret an object as being closer than it actually is on a hot day.

CQ17.3 The speed of sound to two significant figures is 340 m/s. Let's assume that you can measure time to $\frac{1}{10}$ second by using a stopwatch. To get a speed to two significant figures, you need to measure a time of at least 1.0 seconds. Since $d = vt$, the minimum distance is 340 meters.

CQ17.4 When listening, you are approximately the same distance from all of the members of the group. If different frequencies traveled at different speeds, then you might hear the higher pitched frequencies before you heard the lower ones produced at the same time.

CQ17.5 The speed of light is so high that the arrival of the flash is practically simultaneous with the lightning discharge. Thus, the delay between the flash and the arrival of the sound of thunder is the time sound takes to travel the distance separating the lightning from you. By counting the seconds between the flash and thunder and knowing the approximate speed of sound in air, you have a rough measure of the distance to the lightning bolt.

CQ17.6 Both. There are actually two Doppler shifts. The first shift arises from the source (you) moving toward the observer (the cliff). The second arises from the observer (you) moving toward the source (the cliff). If, instead of a cliff, there is a spacecraft moving toward you, then there are shifts due to moving source (you) and moving observer (the spacecraft) before reflection, and moving source (the spacecraft) and moving observer (you) after reflection.

CQ17.7 A beam of radio waves of known frequency is sent toward a speeding car, which reflects the beam back to a detector in the police car. The amount the returning frequency has been shifted depends on the velocity of the oncoming car.

CQ17.8 Our brave Siberian saw the first wave he encountered, light traveling at 3.00×10^8 m/s. At the same moment, infrared as well as visible light began warming his skin, but some time was required to raise the temperature of the outer skin layers before he noticed it. The meteor produced compressional waves in the air and in the ground. The wave in the ground, which can be called either sound or a seismic wave, traveled much faster than the wave in air, since the ground is much stiffer against compression. Our witness received it next and noticed it as a little earthquake. He was no doubt unable to distinguish the P and S waves from each other. The first air-

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compression wave he received was a shock wave with an amplitude on the order of meters. It transported him off his doorstep. Then he could hear some additional direct sound, reflected sound, and perhaps the sound of the falling trees.

CQ17.9 If an object is a half meter from the sonic ranger, then the sensor would have to measure how long it would take for a sound pulse to travel one meter. Because sound of any frequency moves at about 343 m/s, the sonic ranger would have to be able to measure a time difference of under 0.003 seconds. This small time measurement is possible with modern electronics, but it would be more expensive to outfit sonic rangers with the more sensitive equipment than it is to print "do not use to measure distances less than $\frac{1}{2}$ meter" in the users' manual.

SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 17.1 Pressure Variations in Sound Waves

P17.1 (a) $A = \boxed{2.00\ \mu\text{m}}$

(b) $\lambda = \dfrac{2\pi}{15.7} = 0.400\ \text{m} = \boxed{40.0\ \text{cm}}$

(c) $v = \dfrac{\omega}{k} = \dfrac{858}{15.7} = \boxed{54.6\ \text{m/s}}$

(d) $s = 2.00\cos\left[(15.7)(0.050\ 0) - (858)\left(3.00\times10^{-3}\right)\right] = \boxed{-0.433\ \mu\text{m}}$

(e) $v_{\max} = A\omega = (2.00\ \mu\text{m})\left(858\ \text{s}^{-1}\right) = \boxed{1.72\ \text{mm/s}}$

P17.2 (a) $\Delta P = (1.27\ \text{Pa})\sin\left(\dfrac{\pi x}{\text{m}} - \dfrac{340\pi t}{\text{s}}\right)$ (SI units)

The pressure amplitude is: $\Delta P_{\max} = \boxed{1.27\ \text{Pa}}$

(b) $\omega = 2\pi f = 340\pi/\text{s},$ so $f = \boxed{170\ \text{Hz}}$

(c) $k = \dfrac{2\pi}{\lambda} = \pi/\text{m},$ giving $\lambda = \boxed{2.00\ \text{m}}$

(d) $v = \lambda f = (2.00\ \text{m})(170\ \text{Hz}) = \boxed{340\ \text{m/s}}$

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P17.3 We write the pressure variation as $\Delta P = \Delta P_{\max} \sin(kx - \omega t)$. Note that

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{(0.100\ \text{m})} = 62.8\ \text{m}^{-1}$$

and $$\omega = \frac{2\pi v}{\lambda} = \frac{2\pi(343\ \text{m/s})}{(0.100\ \text{m})} = 2.16 \times 10^4\ \text{s}^{-1}.$$

Therefore,

$$\boxed{\Delta P = 0.200 \sin\left[62.8x - 2.16 \times 10^4 t\right]}$$

where ΔP is in Pa, x is in meters, and t is in seconds.

Section 17.2 Speed of Sound Waves

P17.4 We use $\Delta P_{\max} = \rho v \omega s_{\max} = \rho v\left(\frac{2\pi v}{\lambda}\right) s_{\max}$:

$$\lambda_{\min} = \frac{2\pi\rho v^2 s_{\max}}{\Delta P_{\max}} = \frac{2\pi(1.20\ \text{kg/m}^3)(343\ \text{m/s})^2(5.50 \times 10^{-6}\ \text{m})}{0.840\ \text{Pa}} = \boxed{5.81\ \text{m}}$$

***P17.5** $\Delta P_{\max} = \rho\omega v s_{\max} = (1.20\ \text{kg/m}^3)[2\pi(2\,000\ \text{s}^{-1})](343\ \text{m/s})(2.00 \times 10^{-8}\ \text{m})$

$$\Delta P_{\max} = \boxed{0.103\ \text{Pa}}$$

P17.6 The speed of longitudinal waves in a fluid is $v = \sqrt{B/\rho}$. Considering the Earth's crust to consist of a very viscous fluid, our estimate of the average bulk modulus of the material in Earth's crust is

$$B = \rho v^2 = (2\,500\,\text{kg/m}^3)(7 \times 10^3\ \text{m/s})^2 = \boxed{1 \times 10^{11}\ \text{Pa}}$$

P17.7 The sound pulse must travel 150 m before reflection and 150 m after reflection. We have $d = vt$:

$$t = \frac{d}{v} = \frac{300\ \text{m}}{1\,533\ \text{m/s}} = \boxed{0.196\ \text{s}}$$

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P17.8 (a) The speed gradually changes from

$$v = (331\ \text{m/s})\left(1+\frac{27.0^\circ\text{C}}{273^\circ\text{C}}\right)^{1/2} = 347\ \text{m/s}$$

to $$v = (331\ \text{m/s})\left(1+\frac{0^\circ\text{C}}{273^\circ\text{C}}\right)^{1/2} = 331\ \text{m/s}$$

a 4.6% decrease. The cooler air at the same pressure is more dense.

(b) The frequency is unchanged because every wave crest in the hot air becomes one crest without delay in the cold air.

(c) The wavelength decreases by 4.6%, from

$$v/f = (347\ \text{m/s})/(4000/\text{s}) = 86.7\ \text{mm}$$

to $$v/f = (331\ \text{m/s})/(4000/\text{s}) = 82.8\ \text{mm}$$

The crests are more crowded together when they move more slowly.

P17.9 (a) If $f = 2.40$ MHz, $\lambda = \dfrac{v}{f} = \dfrac{1\,500\ \text{m/s}}{2.40\times10^{6}\ \text{s}^{-1}} = \boxed{0.625\ \text{mm}}$

(b) If $f = 1.00$ MHz, $\lambda = \dfrac{v}{f} = \dfrac{1\,500\ \text{m/s}}{10^{6}\ \text{s}^{-1}} = \boxed{1.50\ \text{mm}}$

If $f = 20.0$ MHz, $\lambda = \dfrac{1\,500\ \text{m/s}}{2\times10^{7}\ \text{s}^{-1}} = \boxed{75.0\ \mu\text{m}}$

P17.10 $\Delta P_{\max} = \rho v \omega s_{\max}$

$$s_{\max} = \frac{\Delta P_{\max}}{\rho v \omega} = \frac{4.00\times10^{-3}\ \text{N/m}^2}{(1.20\ \text{kg/m}^3)(343\ \text{m/s})(2\pi)(10.0\times10^{3}\ \text{s}^{-1})}$$

$$= \boxed{1.55\times10^{-10}\ \text{m}}$$

P17.11 (a) Since $v_{\text{light}} >> v_{\text{sound}}$, and assuming that the speed of sound is constant through the air between the lightning strike and the observer, we have

$$d \approx (343\ \text{m/s})(16.2\ \text{s}) = \boxed{5.56\ \text{km}}$$

(b) No, we do not need to know the value of the speed of light. The speed of light is much greater than the speed of sound, so the time interval required for the light to reach you is negligible compared to the time interval for the sound.

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P17.12 It is easiest to solve part (b) first:

(b) The distance the sound travels to the plane is

$$d_s = \sqrt{h^2 + \left(\frac{h}{2}\right)^2} = \frac{h\sqrt{5}}{2}$$

The sound travels this distance in 2.00 s, so

$$d_s = \frac{h\sqrt{5}}{2} = (343\ \text{m/s})(2.00\ \text{s}) = 686\ \text{m}$$

giving the altitude of the plane as $h = \dfrac{2(686\ \text{m})}{\sqrt{5}} = \boxed{614\ \text{m}}$

(a) The distance the plane has traveled in 2.00 s is

$$v(2.00\ \text{s}) = \frac{h}{2} = 307\ \text{m}$$

Thus, the speed of the plane is:

$$v = \frac{307\ \text{m}}{2.00\ \text{s}} = \boxed{153\ \text{m/s}}$$

P17.13 Sound takes this time to reach the man: $\Delta t_s = \dfrac{d-h}{v}$. The minimum time interval between when a warning is shouted and when the man responds to the warning is $\Delta t_{\min} = \Delta t_s + \Delta t$.

Since the whole time interval to fall is given by

$$\Delta y = (d-h) = \frac{1}{2} g \Delta t_f^2 \rightarrow \Delta t_f = \sqrt{\frac{2(d-h)}{g}}$$

The warning needs to come at least

$$\Delta T = \Delta t_f - \Delta t - \Delta t_s = \sqrt{\frac{2(d-h)}{g}} - \Delta t - \frac{d-h}{v}$$

into the fall, when the pot is at the position

$$y_f = y_i + v_{yi}\Delta T - \frac{1}{2} g \Delta T^2$$

$$y_f = 20.0\ \text{m} - \frac{1}{2}\left(9.80\ \text{m/s}^2\right) \times \left(\sqrt{\frac{2(20.0\ \text{m} - 1.75\ \text{m})}{g}} - 0.300\ \text{s} - \frac{20.0\ \text{m} - 1.75\ \text{m}}{343\ \text{m/s}} \right)^2$$

$y_f = \boxed{7.82\ \text{m}}$ above the ground.

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P17.14 Sound takes this time to reach the man: $\Delta t_s = \dfrac{d-h}{v}$. The minimum time interval between when a warning is shouted and when the man responds to the warning is $\Delta t_{\min} = \Delta t_s + \Delta t$.

Since the whole time interval to fall is given by

$$\Delta y = (d-h) = \frac{1}{2} g \Delta t_f^2 \rightarrow \Delta t_f = \sqrt{\frac{2(d-h)}{g}}$$

The warning needs to come at least

$$\Delta T = \Delta t_f - \Delta t - \Delta t_s = \sqrt{\frac{2(d-h)}{g}} - \Delta t - \frac{d-h}{v}$$

into the fall, when the pot is at the position

$$y_f = y_i + v_{yi}\Delta T - \frac{1}{2} g \Delta T^2$$

$$y_f = \boxed{d - \frac{1}{2} g \left(\sqrt{\frac{2(d-h)}{g}} - \Delta t - \frac{d-h}{v} \right)^2} \text{ above the ground.}$$

P17.15 (a) At 9 000 m, $\Delta T = (9\ 000\ \text{m})\left(\dfrac{-1.00°\text{C}}{150\ \text{m}}\right) = -60.0°\text{C}$, so $T = -30.0°\text{C}$.

Using the chain rule,

$$\frac{dv}{dt} = \frac{dv}{dT_C}\frac{dT_C}{dx}\frac{dx}{dt} = v\frac{dv}{dT_C}\frac{dT_C}{dx} = v(0.607)\left(\frac{1}{150}\right) = \frac{v}{247}$$

so $dt = (247\ \text{s})\dfrac{dv}{v}$. Integrating,

$$\int_0^t dt = (247\ \text{s}) \int_{v_i}^{v_f} \frac{dv}{v}$$

$$t = (247\ \text{s}) \ln\left(\frac{v_f}{v_i}\right) = (247\ \text{s}) \ln\left[\frac{331.5 + 0.607(30.0)}{331.5 + 0.607(-30.0)}\right]$$

which gives $t = \boxed{27.2\ \text{s}}$ for sound to reach the ground.

(b) $t = \dfrac{h}{v} = \dfrac{9\ 000\ \text{m}}{331.5\ \text{m/s} + 0.607(30.0°\text{C})} = \boxed{25.7\ \text{s}}$

The time interval in (a) is longer.

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P17.16 Since $\cos^2\theta + \sin^2\theta = 1$, $\sin\theta = \pm\sqrt{1-\cos^2\theta}$ (each sign applying half the time),

$$\Delta P = \Delta P_{\max} \sin(kx - \omega t) = \pm\rho v\omega s_{\max}\sqrt{1-\cos^2(kx-\omega t)}$$

Therefore,

$$\Delta P = \pm\rho v\omega\sqrt{s_{\max}^2 - s_{\max}^2\cos^2(kx-\omega t)} = \pm\rho v\omega\sqrt{s_{\max}^2 - s^2}$$

P17.17 (a) The two pulses travel the same distance, and so the one that travels at the highest velocity will arrive first. Because the speed of sound in air is 343 m/s and the speed of sound in the iron rod is 5 950 m/s,

the pulse travelling through the iron rail will arrive first.

(b) For each of the pulses $t = \dfrac{L}{v}$.

Therefore,

$$t_{\text{rod}} = \frac{L}{v_{\text{rod}}} = \frac{8.50\text{ m}}{5\ 950\text{ m/s}} = 1.43\text{ milliseconds}$$

and $$t_{\text{air}} = \frac{L}{v_{\text{air}}} = \frac{8.50\text{ m}}{343\text{ m/s}} = 24.78\text{ milliseconds}$$

The difference between their two arrival times is

$$\Delta t = t_{\text{air}} - t_{\text{rod}} = 24.78\text{ ms} - 1.43\text{ ms} = \boxed{23.4\text{ ms}}$$

P17.18 Let d_1 represent the cowboy's distance from the nearer canyon wall and d_2 his distance from the farther cliff. The sound for the first echo travels distance $2d_1$. For the second, $2d_2$. For the third, $2d_1 + 2d_2$. For the fourth echo, $2d_1 + 2d_2 + 2d_1$. The time interval between the shot and the first echo is $\Delta t_1 = 2d_1/v$, between the shot and the second echo is $\Delta t_2 = 2d_2/v$, and so on.

Then

$$\Delta t_2 - \Delta t_1 = \frac{2d_2 - 2d_1}{343\text{ m/s}} = 1.92\text{ s and}$$

$$\Delta t_3 - \Delta t_2 = 1.47\text{ s} \quad\rightarrow\quad \frac{(2d_1 + 2d_2) - 2d_2}{343\text{ m/s}} = \frac{2d_1}{343\text{ m/s}} = 1.47\text{ s}$$

Thus, $$d_1 = \frac{1}{2}(343\text{ m/s})(1.47\text{ s}) = 252\text{ m}, \text{ and } \Delta t_1 = 1.47\text{ s}$$

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From above,

$$\Delta t_2 - \Delta t_1 = 1.92 \text{ s} \quad \rightarrow \quad \frac{2d_2}{343 \text{ m/s}} = 1.92 \text{ s} + 1.47 \text{ s}$$

which gives $d_2 = 581 \text{ m}$

(a) So, $d_1 + d_2 = \boxed{833 \text{ m}}$.

(b) $$\frac{2d_1 + 2d_2 + 2d_1 - \left(2d_1 + 2d_2\right)}{343 \text{ m/s}} = \frac{2d_2}{343 \text{ m/s}} = \boxed{1.47 \text{ s}}$$

Section 17.3 Intensity of Periodic Sound Waves

P17.19 We use Equation 17.14:

$$\beta = (10 \text{ dB}) \log\left(\frac{I}{I_0}\right) = (10 \text{ dB}) \log\left(\frac{4.00 \times 10^{-6} \text{ W/m}^2}{1.00 \times 10^{-12} \text{ W/m}^2}\right)$$

$$= \boxed{66.0 \text{ dB}}$$

P17.20 The sound power incident on the eardrum is $P = IA$, where I is the intensity of the sound and $A = 5.00 \times 10^{-5} \text{ m}^2$ is the area of the eardrum.

(a) At the threshold of pain, $I = 1.00 \text{ W/m}^2$.

Thus, $P = IA = \left(5.00 \times 10^{-5} \text{ m}^2\right)\left(1.00 \text{ W/m}^2\right) = \boxed{5.00 \times 10^{-5} \text{ W}}$

(b) Energy transfer can be obtained from power by $P = \dfrac{E}{\Delta t} \rightarrow E = P\Delta t$. Thus,

$$E = P\Delta t = \left(5.00 \times 10^{-5} \text{ J/s}\right)(60.0 \text{ s}) = \boxed{3.00 \times 10^{-3} \text{ J}}$$

P17.21 We use $I = \dfrac{1}{2}\rho\omega^2 s_{\max}^2 v$.

(a) At $f = 2\,500$ Hz, the frequency is increased by a factor of 2.50, so the intensity (at constant $s_{\max}$) increases by $(2.50)^2 = 6.25$.

Therefore, $6.25\left(0.600\right) = \boxed{3.75 \text{ W/m}^2}$

(b) The changes cancel each other: frequency $f \rightarrow f' = f/2$, and displacement amplitude $s_{\max} \rightarrow s'_{\max} = 2s_{\max}$

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originial intensity: $I = \frac{1}{2}\rho\omega^2 s_{\max}^2 v = 0.600 \text{ W/m}^2$

new intensity: $I' = \frac{1}{2}\rho\omega'^2 s_{\max}'^2 v = \frac{1}{2}\rho\left(\frac{\omega}{2}\right)^2 \left(2s_{\max}\right)^2 v = \frac{1}{2}\rho\omega^2 s_{\max}^2 v$

$$= \boxed{600 \text{ W/m}^2}$$

P17.22 The original intensity is $I_1 = \frac{1}{2}\rho\omega^2 s_{\max}^2 v = 2\pi^2 \rho v f^2 s_{\max}^2$

(a) If the frequency is increased to f' while a constant displacement amplitude is maintained, the new intensity is

$$I_2 = 2\pi^2 \rho v \left(f'\right)^2 s_{\max}^2 \text{ so } \frac{I_2}{I_1} = \frac{2\pi^2 \rho v \left(f'\right) s_{\max}^2}{2\pi^2 \rho v f^2 s_{\max}^2} = \left(\frac{f'}{f}\right)^2$$

or $\boxed{I_2 = \left(\frac{f'}{f}\right)^2 I_1}$

(b) If the frequency is reduced to $f' = \frac{f}{2}$ while the displacement amplitude is doubled, the new intensity is

$$I_2 = 2\pi^2 \rho v \left(\frac{f}{2}\right)^2 \left(2s_{\max}\right)^2 = 2\pi^2 \rho v f^2 s_{\max}^2 = I_1$$

or the $\boxed{\text{intensity is unchanged}}$.

P17.23 In terms of their intensities, the difference in the decibel level of two sounds is

$$\beta_2 - \beta_1 = (10 \text{ dB}) \log\left(\frac{I_2}{I_0}\right) - (10 \text{ dB}) \log\left(\frac{I_1}{I_0}\right)$$

$$= (10 \text{ dB}) \log\left(\frac{I_2}{\cancel{I_0}} \cdot \frac{\cancel{I_0}}{I_1}\right) = (10 \text{ dB}) \log\left(\frac{I_2}{I_1}\right)$$

Thus, $\frac{I_2}{I_1} = 10^{(\beta_2 - \beta_1)/10}$ or $I_2 = I_1 \times 10^{(\beta_2 - \beta_1)/10}$

If $\beta_2 - \beta_1 = 30 \text{ dB}$ and $I_1 = 3.0 \times 10^{-11} \text{ W/m}^2$, then

$$I_2 = \left(3.0 \times 10^{-11} \text{ W/m}^2\right) \times 10^3 = \boxed{3.0 \times 10^{-8} \text{ W/m}^2}$$

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P17.24 The intensity is given by $I = \dfrac{P_{\text{avg}}}{4\pi r^2}$.

The power is not given, but the intensity at a known distance is $I = \dfrac{P_{\text{avg}}}{4\pi r^2}$, which gives

$$P_{\text{avg}} = I(r)4\pi r^2 = 4\pi\left(0.25\ \text{W/m}^2\right)\left(16\ \text{m}\right)^2 = 804.2\ \text{W}$$

which can then be substituted back into the same equation:

$$I = \frac{P_{\text{avg}}}{4\pi r^2} = \frac{804.2\ \text{W}}{4\pi\left(28\ \text{m}\right)^2} = \boxed{0.082\ \text{W/m}^2}$$

P17.25 (a) From the sound level equation,

$$120\ \text{dB} = \left(10\ \text{dB}\right)\log\left[\frac{I}{10^{-12}\ \text{W/m}^2}\right]$$

$$I = 1.00\ \text{W/m}^2 = \frac{P}{4\pi r^2}$$

$$r = \sqrt{\frac{P}{4\pi I}} = \sqrt{\frac{6.00\ \text{W}}{4\pi\left(1.00\ \text{W/m}^2\right)}} = \boxed{0.691\ \text{m}}$$

We have assumed the speaker is an isotropic point source.

(b) Again from the sound level equation,

$$0\ \text{dB} = \left(10\ \text{dB}\right)\log\left(\frac{I}{10^{-12}\ \text{W/m}^2}\right)$$

$$I = 1.00\times10^{-12}\ \text{W/m}^2$$

$$r = \sqrt{\frac{P}{4\pi I}} = \sqrt{\frac{6.00\ \text{W}}{4\pi\left(1.00\times10^{-12}\ \text{W/m}^2\right)}} = \boxed{691\,\text{km}}$$

We have assumed a uniform medium that absorbs no energy.

P17.26 The decibel level due to the first siren is

$$\beta_1 = \left(10\ \text{dB}\right)\log\left(\frac{100.0\ \text{W/m}^2}{1.0\times10^{-12}\ \text{W/m}^2}\right) = 140\ \text{dB}$$

Thus, the decibel level of the sound from the ambulance is

$$\beta_2 = \beta_1 + 10\ \text{dB} = 140\ \text{dB} + 10\ \text{dB} = \boxed{150\ \text{dB}}$$

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***P17.27** (a) The intensity of sound at 10 km from the horn (where $\beta = 50$ dB) is

$$I = I_0 10^{\beta/10} = \left(1.0\times10^{-12}\ \text{W/m}^2\right)10^{5.0} = 1.0\times10^{-7}\ \text{W/m}^2$$

Thus, from $I = \dfrac{P}{4\pi r^2}$, the power emitted by the source is

$$P = 4\pi r^2 I = 4\pi\left(10.0\times10^{3}\ \text{m}\right)^2\left(1.0\times10^{-7}\ \text{W/m}^2\right) = 126\ \text{W}$$

(b) At $r = 50$ m, the intensity of the sound will be

$$I = \frac{P}{4\pi r^2} = \frac{1.3\times10^{2}\ \text{W}}{4\pi(50\ \text{m})^2} = 4.0\times10^{-3}\ \text{W/m}^2$$

and the sound level is

$$\beta = (10\ \text{dB})\log\left(\frac{I}{I_0}\right) = (10\ \text{dB})\log\left(\frac{4.0\times10^{-3}\ \text{W/m}^2}{1.0\times10^{-12}\ \text{W/m}^2}\right)$$

$$= \boxed{96\ \text{dB}}$$

P17.28 (a) The sound intensity inside the church is given by

$$\beta = (10\ \text{dB})\log\left(\frac{I}{I_0}\right)$$

$$101\ \text{dB} = (10\ \text{dB})\log\left(\frac{I}{10^{-12}\ \text{W/m}^2}\right)$$

$$I = 10^{10.1}\left(10^{-12}\ \text{W/m}^2\right) = 10^{-1.90}\ \text{W/m}^2 = 0.012\ 6\ \text{W/m}^2$$

We suppose that sound comes perpendicularly out through the windows and doors. Then, the radiated power is

$$P = IA = \left(0.012\ 6\ \text{W/m}^2\right)\left(22.0\ \text{m}^2\right) = 0.277\ \text{W}$$

Are you surprised by how small this is? The energy radiated in 20.0 minutes is

$$E = Pt = (0.277\ \text{J/s})(20.0\ \text{min})\left(\frac{60.0\ \text{s}}{1.00\ \text{min}}\right) = \boxed{332\ \text{J}}$$

(b) If the ground reflects all sound energy headed downward, the sound power, $P = 0.277$ W, covers the area of a hemisphere. One kilometer away, this area is

$$A = 2\pi r^2 = 2\pi(1\,000\ \text{m})^2 = 2\pi\times10^{6}\ \text{m}^2$$

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The intensity at this distance is

$$I = \frac{P}{A} = \frac{0.277\ \text{W}}{2\pi \times 10^{6}\ \text{m}^2} = 4.41 \times 10^{-8}\ \text{W/m}^2$$

and the sound intensity level is

$$\beta = (10\ \text{dB}) \log\left(\frac{4.41 \times 10^{-8}\ \text{W/m}^2}{1.00 \times 10^{-12}\ \text{W/m}^2} \right) = \boxed{46.4\ \text{dB}}$$

P17.29 (a) For the initial low note the wavelength is

$$\lambda = \frac{v}{f} = \frac{343\ \text{m/s}}{146.8\ /\ \text{s}} = \boxed{2.34\ \text{m}}$$

(b) For the final high note $\lambda = \dfrac{343\ \text{m/s}}{880\ /\ \text{s}} = \boxed{0.390\ \text{m}}$

We observe that the ratio of the frequencies of these two notes is $\dfrac{880\ \text{Hz}}{146.8\ \text{Hz}} = 5.99,$ nearly equal to a small integer. This fact is associated with the consonance of the notes D and A.

(c, d) The intensity level for both notes is the same 75.0 dB:

$$\beta = (10\ \text{dB}) \log\left(\frac{I}{10^{-12}\ \text{W/m}^2} \right) = 75\ \text{dB}$$

gives $I = 3.16 \times 10^{-5}\ \text{W/m}^2$

Therefore, the pressure amplitude for both low and high notes is the same, and $I = \dfrac{\Delta P_{\max}^2}{2\rho v}$ gives

$$\begin{aligned} \Delta P_{\max} &= \sqrt{2\rho v I} \\ &= \sqrt{2\left(1.20\ \text{kg/m}^3\right)\left(343\ \text{m/s}\right)\left(3.16 \times 10^{-5}\ \text{W/m}^2\right)} \\ &= \boxed{0.161\ \text{Pa}} \end{aligned}$$

(e) $$I = \frac{1}{2}\rho v(\omega s_{\max})^2 = \frac{1}{2}\rho v 4\pi^2 f^2 s_{\max}^2 \ \rightarrow\ s_{\max} = \sqrt{\frac{I}{2\pi^2 \rho v f^2}} = \frac{1}{f}\sqrt{\frac{I}{2\pi^2 \rho v}}$$

We see that for the same intensity level, the displacement amplitude is inversely proportional to the frequency.

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For the low note,

$$s_{\max} = \frac{1}{146.8/\text{s}}\sqrt{\frac{3.16\times10^{-5}\ \text{W/m}^2}{2\pi^2\,(1.20\ \text{kg/m}^3)(343\ \text{m/s})}}$$

$$= \frac{6.24\times10^{-5}\ \text{m/s}}{146.8\ \text{s}^{-1}} = \boxed{4.25\times10^{-7}\ \text{m}}$$

(f) For the high note,

$$s_{\max} = \frac{6.24\times10^{-5}\ \text{m/s}}{880\ \text{s}^{-1}} = \boxed{7.09\times10^{-8}\ \text{m}}$$

P17.30 We begin with $\beta_2 = (10\ \text{dB})\log\left(\frac{I_2}{I_0}\right)$ and $\beta_1 = (10\ \text{dB})\log\left(\frac{I_1}{I_0}\right)$, so

$$\beta_2 - \beta_1 = (10\ \text{dB})\log\left(\frac{I_2}{I_1}\right)$$

Also, $I_2 = \frac{P}{4\pi r_2^2}$ and $I_1 = \frac{P}{4\pi r_1^2}$, giving $\frac{I_2}{I_1} = \left(\frac{r_1}{r_2}\right)^2$

Then, $\beta_2 - \beta_1 = (10\ \text{dB})\log\left(\frac{r_1}{r_2}\right)^2 = \boxed{20\log\left(\frac{r_1}{r_2}\right)}$

P17.31 From $\beta = 10\ \log\left(\frac{I}{10^{-12}\ \text{W/m}^2}\right)$, we have

$$I = [10^{\beta/10}]\left(10^{-12}\ \text{W/m}^2\right)$$

(a) For your baby,

$$I_b = \left(10^{75.0/10}\right)\left(10^{-12}\ \text{W/m}^2\right) = 3.16\times10^{-5}\ \text{W/m}^2$$

For the music,

$$I_m = \left(10^{80.0/10}\right)\left(10^{-12}\ \text{W/m}^2\right) = 10.0\times10^{-5}\ \text{W/m}^2$$

The combined intensity is

$$I_{\text{total}} = I_m + I_b$$

$$= 10.0\times10^{-5}\ \text{W/m}^2 + 3.16\times10^{-5}\ \text{W/m}^2$$

$$= \boxed{13.2\times10^{-5}\ \text{W/m}^2}$$

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(b) The combined sound level is then

$$\beta_{\text{total}} = 10\log\left(\frac{I_{\text{total}}}{10^{-12}\ \text{W/m}^2}\right) = 10\log\left(\frac{1.32\times10^{-4}\ \text{W/m}^2}{10^{-12}\ \text{W/m}^2}\right)$$

$$= \boxed{81.2\ \text{dB}}$$

P17.32 The speakers broadcast equally in all directions, so the intensity of sound is inversely proportional to the square of the distance from its source.

(a) $r_{AC} = \sqrt{3.00^2 + 4.00^2}\ \text{m} = 5.00\ \text{m}$

$$I = \frac{P}{4\pi r^2} = \frac{1.00\times10^{-3}\ \text{W}}{4\pi\left(5.00\ \text{m}\right)^2} = 3.18\times10^{-6}\ \text{W/m}^2$$

$$\beta = \left(10\ \text{dB}\right)\log\left(\frac{3.18\times10^{-6}\ \text{W/m}^2}{10^{-12}\ \text{W/m}^2}\right)$$

$$\beta = \left(10\ \text{dB}\right)\ 6.50 = \boxed{65.0\,\text{dB}}$$

(b) $r_{BC} = 4.47\ \text{m}$

$$I = \frac{1.50\times10^{-3}\ \text{W}}{4\pi\left(4.47\ \text{m}\right)^2} = 5.97\times10^{-6}\ \text{W/m}^2$$

$$\beta = (10\ \text{dB})\log\left(\frac{5.97\times10^{-6}\ \text{W/m}^2}{10^{-12}\ \text{W/m}^2}\right)$$

$$\beta = \boxed{67.8\ \text{dB}}$$

(c) $I = 3.18\ \mu\text{W/m}^2 + 5.97\ \mu\text{W/m}^2$

$$\beta = (10\ \text{dB})\log\left(\frac{9.15\times10^{-6}\ \text{W/m}^2}{10^{-12}\ \text{W/m}^2}\right) = \boxed{69.6\ \text{dB}}$$

P17.33 The sound intensity at distance d_1 is, suppressing units,

$$I_1 = \frac{\Delta P_{\max}^2}{2\rho v} = \frac{(10.0)^2}{2(1.20)(343)} = 0.121\ \text{W/m}^2$$

If air does not absorb sound energy, the intensity of sound is inversely proportional to the square of the distance from its source. The intensity at distance d_2 is

$$I_2 = \left(\frac{d_1}{d_2}\right)^2 I_1 = \left(\frac{500\ \text{m}}{4\ 000\ \text{m}}\right)^2 I_1 = \frac{1}{64}\left(0.121\ \text{W/m}^2\right)$$

$$= 1.89\times10^{-3}\ \text{W/m}^2$$

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which has an intensity level of

$$\beta_2 = (10\text{ dB})\log\left(\frac{I_2}{I_0}\right) = (10\text{ dB})\log\left(\frac{1.89\times10^{-3}\text{ W/m}^2}{10^{-12}\text{ W/m}^2}\right)$$
$$= 92.77\text{ dB}$$

Allowing for absorption of the wave over the distance traveled,

$$\beta_2' = \beta_2 - (7.00\text{ dB/km})(3.50\text{ km}) = \boxed{68.3\text{ dB}}$$

This is equivalent to the sound intensity level of heavy traffic.

P17.34 (a) The energy transferred by sound from the explosion is

$$T_{MW} = P\Delta t = 4\pi r^2 I\Delta t$$
$$= 4\pi(100\text{ m})^2\left(7.00\times10^{-2}\text{ W/m}^2\right)(0.200\text{ s})$$
$$= \boxed{1.76\text{ kJ}}$$

(b) $$\beta = (10\text{ dB})\log\left(\frac{7.00\times10^{-2}}{1.00\times10^{-12}}\right) = \boxed{108\text{ dB}}$$

P17.35 From the definition of sound level,

$$\beta = 10\log\left(\frac{I}{10^{-12}\text{ W/m}^2}\right)$$

we can compute the intensities corresponding to each of the levels mentioned as $I = [10^{\beta/10}]10^{-12}\text{ W/m}^2$.

They are $I_{120} = 1\text{ W/m}^2$

$I_{100} = 10^{-2}\text{ W/m}^2$

and $I_{10} = 10^{-11}\text{ W/m}^2$

(a) The power passing through any sphere around the source is $Power = 4\pi r^2 I$. If we ignore absorption of sound by the medium, conservation of energy for the sound wave as a system requires that $r_{120}^2 I_{120} = r_{100}^2 I_{100} = r_{10}^2 I_{10}$. Then

$$r_{100} = r_{120}\sqrt{\frac{I_{120}}{I_{100}}} = (3.00\text{ m})\sqrt{\frac{1\text{ W/m}^2}{10^{-2}\text{ W/m}^2}} = \boxed{30.0\text{ m}}$$

(b) $$r_{10} = r_{120}\sqrt{\frac{I_{120}}{I_{10}}} = (3.00\text{ m})\sqrt{\frac{1\text{ W/m}^2}{10^{-11}\text{ W/m}^2}} = \boxed{9.49\times10^5\text{ m}}$$

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P17.36 We assume that both lawn mowers are equally loud and approximately the same distance away. We found in Example 17.3 that a sound of twice the intensity results in an increase in sound level of 3 dB. We also see from the What If? section of that example that a doubling of loudness requires a 10-dB increase in sound level. Therefore, the sound of two lawn mowers will not be twice the loudness, but only a little louder than one!

Section 17.4 The Doppler Effect

P17.37 The source and detector of waves are both moving with respect to the medium in which the waves are travelling.

(a) The general form of the Doppler equation is:

$$f' = \left(\frac{v + v_o}{v - v_s} \right) f$$

where the positive signs for v_o and v_s are for source or observer approaching each other. When the ambulance is approaching the car from behind:

$$f' = \left(\frac{v + v_O}{v - v_S} \right) f = \left(\frac{343 + (-25)}{343 - (+42)} \right) 450 \text{ Hz} = (1.056)450 \text{ Hz}$$

$$= \boxed{475 \text{ Hz}}$$

(b) When the ambulance is moving away in front of the moving car:

$$f' = \left(\frac{v + v_O}{v - v_S} \right) f = \left(\frac{343 + (+25)}{343 - (-42)} \right) 450 \text{ Hz} = (0.956)450 \text{ Hz}$$

$$= \boxed{430 \text{ Hz}}$$

P17.38 The *half angle* of the shock wave cone is given by $\sin\theta = \dfrac{v_{\text{light}}}{v_S}$.

$$v_S = \frac{v_{\text{light}}}{\sin\theta} = \frac{2.25 \times 10^8 \text{ m/s}}{\sin(53.0°)} = \boxed{2.82 \times 10^8 \text{ m/s}}$$

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P17.39 (a) The Doppler-shifted frequency is found from

$$f' = \frac{f\left(v + v_o\right)}{\left(v - v_s\right)}$$

$$= \left(2\,500\text{ Hz}\right)\frac{(343 + 25.0)}{(343 - 40.0)} = \boxed{3.04\text{ kHz}}$$

(b) After the police car passes,

$$f' = \left(2\,500\text{ Hz}\right)\left(\frac{343 + (-25.0)}{343 - (-40.0)}\right) = \boxed{2.08\text{ kHz}}$$

(c) While the police car overtakes the driver,

$$f' = \left(2\,500\text{ Hz}\right)\left(\frac{343 + (-25.0)}{343 - 40.0}\right) = \boxed{2.62\text{ kHz}}$$

After the police car passes,

$$f' = \left(2\,500\text{ Hz}\right)\left(\frac{343 + 25.0}{343 - (-40.0)}\right) = \boxed{2.40\text{ kHz}}$$

P17.40 (a) Equation 17.19, $f' = f\left(\frac{v + v_o}{v - v_s}\right)$, applies to an observer on $\boxed{\text{B}}$ because B is receiving sound from source A.

(b) The sign of v_s should be $\boxed{\text{positive}}$ because the source is moving toward the observer, resulting in an increase in frequency.

(c) The sign of v_o should be $\boxed{\text{negative}}$ because the observer is moving away from the source, resulting in a decrease in frequency.

(d) The speed of sound should be that of the medium of seawater, $\boxed{1\,533\text{ m/s}}$.

(e) $$f_o = f_s\left(\frac{v + v_o}{v - v_s}\right) = \left(5.27 \times 10^3\text{ Hz}\right)\left[\frac{(1\,533\text{ m/s}) + (-3.00\text{ m/s})}{(1\,533\text{ m/s}) - (+11.0\text{ m/s})}\right]$$

$$= \boxed{5.30 \times 10^3\text{ Hz}}$$

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P17.41 (a) The maximum speed of the speaker is described by

$$\frac{1}{2}mv_{\max}^2 = \frac{1}{2}kA^2$$

$$v_{\max} = \sqrt{\frac{k}{m}}A = \sqrt{\frac{20.0\ \text{N/m}}{5.00\ \text{kg}}}\left(0.500\ \text{m}\right) = 1.00\ \text{m/s}$$

The frequencies heard by the stationary observer range from

$$f'_{\max} = f\left(\frac{v}{v - v_{\max}}\right) = 440\ \text{Hz}\left(\frac{343}{343 - 1.00}\right) = \boxed{441\ \text{Hz}}$$

to

(b) $$f'_{\min} = f\left(\frac{v}{v + v_{\max}}\right) = 440\ \text{Hz}\left(\frac{343}{343 + 1.00}\right) = \boxed{439\ \text{Hz}}$$

(c) $$\beta = (10\ \text{dB})\log\left(\frac{I}{I_0}\right) = (10\ \text{dB})\log\left(\frac{P/4\pi r^2}{I_0}\right)$$

The maximum intensity level $\beta_{\max} = 60.0$ dB occurs at $r = r_{\min} = 1.00$ m. The minimum intensity level occurs when the speaker is farthest from the listener, i.e., when $r = r_{\max} = r_{\min} + 2A = 2.00$ m.

Thus, $$\beta_{\max} - \beta_{\min} = (10\ \text{dB})\log\left(\frac{P}{4\pi I_0 r_{\min}^2}\right) - (10\ \text{dB})\log\left(\frac{P}{4\pi I_0 r_{\max}^2}\right)$$

or

$$\beta_{\max} - \beta_{\min} = (10\ \text{dB})\log\left(\frac{P}{4\pi I_0 r_{\min}^2}\frac{4\pi I_0 r_{\max}^2}{P}\right)$$

$$= (10\ \text{dB})\log\left(\frac{r_{\max}^2}{r_{\min}^2}\right) = (20\ \text{dB})\log\left(\frac{r_{\max}}{r_{\min}}\right)$$

This gives:

$$60.0\ \text{dB} - \beta_{\min} = (20\ \text{dB})\log(2.00) = 6.02\ \text{dB}$$

or $$\beta_{\min} = \boxed{54.0\ \text{dB}}$$

P17.42 The maximum speed of the speaker is described by

$$\frac{1}{2}mv_{\max}^2 = \frac{1}{2}kA^2$$

$$v_{\max} = \sqrt{\frac{k}{m}}A$$

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The frequencies heard by the stationary observer range from

(a) $f'_{\max} = \boxed{\dfrac{vf}{v - A\sqrt{\dfrac{k}{m}}}}$ to (b) $f'_{\max} = \boxed{\dfrac{vf}{v + A\sqrt{\dfrac{k}{m}}}}$

where v is the speed of sound.

(c) $$\beta = (10\text{ dB})\log\left(\frac{I}{I_0}\right) = (10\text{ dB})\log\left(\frac{P/4\pi r^2}{I_0}\right)$$

The maximum intensity level $\beta_{\max} = \beta$ occurs at $r = r_{\min} = d$. The minimum intensity level occurs when the speaker is farthest from the listener, i.e., when $r = r_{\max} = r_{\min} + 2A = d + 2A$.

Thus,

$$\beta_{\max} - \beta_{\min} = (10\text{ dB})\log\left(\frac{P}{4\pi I_0 r_{\min}^2}\right) - (10\text{ dB})\log\left(\frac{P}{4\pi I_0 r_{\max}^2}\right)$$

or

$$\beta_{\max} - \beta_{\min} = (10\text{ dB})\log\left(\frac{P}{4\pi I_0 r_{\min}^2}\frac{4\pi I_0 r_{\max}^2}{P}\right)$$

$$= (10\text{ dB})\log\left(\frac{r_{\max}^2}{r_{\min}^2}\right) = (20\text{ dB})\log\left(\frac{r_{\max}}{r_{\min}}\right)$$

This gives:

$$\beta - \beta_{\min} = (20\text{ dB})\log\left(\frac{d + 2A}{d}\right)$$

or $$\beta_{\min} = \boxed{\beta - (20\text{ db})\log\left(1 + \frac{2A}{d}\right)}$$

P17.43 (a) $$\omega = 2\pi f = 2\pi\left(\frac{115\text{ min}^{-1}}{60.0\text{ s/min}}\right) = 12.0\text{ rad/s}$$

$$v_{\max} = \omega A = (12.0\text{ rad/s})\left(1.80 \times 10^{-3}\text{ m}\right) = \boxed{0.021\,7\text{ m/s}}$$

(b) The heart wall is a moving observer:

$$\Delta f' = f' - f = f\left(\frac{v + v_O}{v} - f\right) - f = f\left(\frac{v_O}{v}\right)$$

$$= (2\,000\,000\text{ Hz})\left(\frac{0.021\,7}{1\,500}\right) = \boxed{28.9\text{ Hz}}$$

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(c) Now, the heart wall is a moving source:

$$\Delta f'' = f'\left(\frac{v}{v - v_s}\right) - f = f\left(\frac{v + v_o}{v}\right)\left(\frac{v}{v - v_s}\right) - f$$

$$\Delta f'' = f\left[\left(\frac{v\left(v + v_o\right)}{v\left(v - v_s\right)}\right) - \left(\frac{v\left(v - v_s\right)}{v\left(v - v_s\right)}\right)\right] = f\left[\frac{v_o + v_s}{v - v_s}\right]$$

Since the velocities of the source and the observer in these expressions are both referring to the movement of the heart wall, and the velocity of the sound wave is much greater than those velocities, we may approximate:

$$\Delta f'' \cong f\left[\frac{2v_s}{v}\right]$$

$$\Delta f'' \cong \left(2.00 \times 10^6 \text{ Hz}\right)\left[\frac{0.043\ 4}{1\ 500}\right] = \boxed{57.9 \text{ Hz}}$$

P17.44 The apparent frequency drops because of the Doppler effect. Using a T subscript for the situation when the athlete moves *toward* the horn, and A for movement away from the horn, we have,

$$\frac{f'_A}{f'_T} = \frac{\left(\frac{v + v_{OA}}{v - v_S}\right)f}{\left(\frac{v + v_{OT}}{v - v_S}\right)f} = \frac{v + v_{OA}}{v + v_{OT}} = \frac{v + \left(-v_O\right)}{v + \left(+v_O\right)} = \frac{v - v_O}{v + v_O}$$

where v_0 is the constant speed of the athlete. Setting this ratio equal to 5/6, we have

$$\frac{5}{6} = \frac{v - v_O}{v + v_O} \quad \rightarrow \quad 5v + 5v_O = 6v - 6v_O \quad \rightarrow \quad 11v_O = v$$

Solving for the speed of the athlete,

$$v_O = \frac{v}{11} = \frac{343 \text{ m/s}}{11} = 31.2 \text{ m/s}$$

This is much faster than a human athlete can run.

P17.45 Let v_a represent the magnitude of the velocity of the ambulance.

As it approaches you hear the frequency $f' = v\left(\frac{v}{v - v_a}\right)f = 560$ Hz.

The negative sign appears because the source is moving toward the observer. The opposite sign with source velocity magnitude describes the ambulance moving away. As the ambulance recedes, the Doppler-

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shifted frequency is

$$f'' = \left(\frac{v}{v + v_a}\right) f = 480 \text{ Hz}.$$

Solving the second of these equations for f and substituting into the other gives

$$f' = \left(\frac{v}{v - v_a}\right)\left(\frac{v + v_a}{v}\right) f'' \quad \text{or} \quad f'v - f'v_a = vf'' + v_a f''$$

so the speed of the source is

$$v_a = \frac{v(f' - f'')}{f' + f''} = \frac{(343 \text{ m/s})(560 \text{ Hz} - 480 \text{ Hz})}{560 \text{ Hz} + 480 \text{ Hz}} = \boxed{26.4 \text{ m/s}}$$

P17.46 We first determine how fast the tuning fork is falling to emit sound with apparent frequency 485 Hz. Call the magnitude of its velocity v_{fall}. The tuning fork source is moving away from the listener, so $v_s = -v_{\text{fall}}$.

Therefore, we use the equation $$f' = \left(\frac{v}{v + v_{\text{fall}}}\right) f$$

Solving for v_{fall} gives $\dfrac{v + v_{\text{fall}}}{v} = \dfrac{f}{f'}$ and $v_{\text{fall}} = v\left(\dfrac{f}{f'} - 1\right)$.

Substituting, we have $v_{\text{fall}} = \left(\dfrac{512 \text{ Hz}}{485 \text{ Hz}} - 1\right)(343 \text{ m/s}) = 19.1 \text{ m/s}.$

The time interval required for the tuning fork to reach this speed, from the particle under constant acceleration model, is given by

$$v_y = 0 + a_y t \quad \text{as} \quad t = v_y/a_y = (19.1 \text{ m/s})/(9.80 \text{ m/s}^2) = 1.95 \text{ s}$$

The distance that the fork has fallen is

$$\Delta y = 0 + \frac{1}{2} a_y t^2 = \frac{1}{2}(9.80 \text{ m/s}^2)(1.95 \text{ s})^2 = 18.6 \text{ m}$$

At this moment, the fork would appear to ring at 485 Hz to a stationary observer just above the fork. However, some additional time is required for the waves to reach the point of release. The fork is moving down, but the sound it radiates still travels away from its instantaneous position at 343 m/s. From the traveling wave model, the time interval it takes to return to the listener is

$$\Delta t = \Delta y/v = 18.6 \text{ m}/(343 \text{ m/s}) = 0.054\,2 \text{ s}$$

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Over a total time $t + \Delta t = 1.95\text{ s} + 0.054\,2\text{ s} = 2.00\text{ s}$, the fork falls a total distance

$$d_{\text{total}} = \frac{1}{2} g t_{\text{total fall}}^2 = \boxed{19.7\text{ m}}$$

P17.47 (a) We find the shock angle from

$$\theta = \sin\left(\frac{v}{v_S}\right)^{-1} = \sin\left(\frac{1}{3.00}\right)^{-1} = 19.5°$$

from $\tan\theta = \dfrac{h}{x}$,

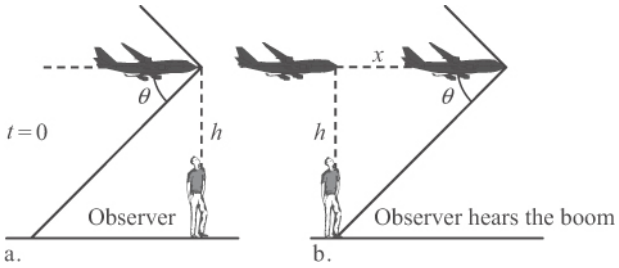
$$x = \frac{h}{\tan\theta} = \frac{20\,000\text{ m}}{\tan 19.5°} = 5.66 \times 10^4\text{ m} = 56.6\text{ km}$$

It takes the plane

$$t = \frac{x}{v_S} = \frac{5.66 \times 10^4\text{ m}}{3.00(335\text{ m/s})} = \boxed{56.3\text{ s}}$$

to travel this distance.

(b) From part (a), $x = \boxed{56.6\text{ km}}$



ANS. FIG. P17.47

Additional Problems

***P17.48** The size of the insect detected by the bat will be comparable to the wavelength of sound emitted by the bat:

$$\lambda = \frac{v}{f} = \frac{340\text{ m/s}}{60.0 \times 10^3\text{ s}^{-1}} = \boxed{5.67\text{ mm}}$$

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***17.49** At normal body temperature of $T = 37.0°\text{C}$, the speed of sound in air is

$$v = (331\ \text{m/s})\sqrt{1+\frac{T_C}{273}} = (331\ \text{m/s})\sqrt{1+\frac{37.0}{273}} = 353\ \text{m/s}$$

and the wavelength of sound having a frequency of $f = 20\,000$ Hz is

$$\lambda = \frac{v}{f} = \frac{353\ \text{m/s}}{20\,000\ \text{Hz}} = 1.76\times10^{-2}\ \text{m} = \boxed{1.76\ \text{cm}}$$

Thus, the diameter of the eardrum is $\boxed{1.76\ \text{cm.}}$

P17.50 (a) The wavelength of the note is

$$\lambda = \frac{v}{f} = \frac{343\ \text{m/s}}{1\,480\ \text{s}^{-1}} = \boxed{0.232\ \text{m}}$$

(b) We find the intensity of the 81.0 dB sound from

$$\beta = 81.0\ \text{dB} = (10\ \text{dB})\log\left[\frac{I}{10^{-12}\ \text{W/m}^2}\right]$$

Then,

$$I = \left(10^{-12}\ \text{W/m}^2\right)10^{8.10} = 10^{-3.90}\ \text{W/m}^2 = 1.26\times10^{-4}\ \text{W/m}^2$$

$$= \frac{1}{2}\rho v\omega^2 s_{\max}^2$$

Which gives a displacement amplitude of

$$s_{\max} = \sqrt{\frac{2I}{\rho v\omega^2}} = \sqrt{\frac{2\left(1.26\times10^{-4}\ \text{W/m}^2\right)}{\left(1.20\ \text{kg/m}^3\right)\left(343\ \text{m/s}\right)4\pi^2\left(1\,480\ \text{s}^{-1}\right)^2}}$$

$$= \boxed{8.41\times10^{-8}\ \text{m}}$$

(c) The wavelength of the F above high C is

$$\lambda' = \frac{v}{f'} = \frac{343\ \text{m/s}}{1\,397\ \text{s}^{-1}} = 0.246\ \text{m}$$

and the change in wavelength is

$$\Delta\lambda = \lambda' - \lambda = 0.246\ \text{m} - 0.232\ \text{m} = \boxed{13.8\ \text{mm}}$$

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P17.51 The trucks form a train analogous to a wave train of crests with speed $v = 19.7$ m/s and unshifted frequency $f = \dfrac{2}{3.00 \text{ min}} = 0.667 \text{ min}^{-1}$.

(a) The cyclist as observer measures a lower Doppler-shifted frequency:

$$f' = f\left(\frac{v + v_o}{v}\right) = \left(0.667 \text{ min}^{-1}\right)\left(\frac{19.7 + (-4.47)}{19.7}\right)$$

$$= \boxed{0.515/\text{min}}$$

(b) $$f'' = f\left(\frac{v + v_o'}{v}\right) = \left(0.667 \text{ min}^{-1}\right)\left(\frac{19.7 + (-1.56)}{19.7}\right) = \boxed{0.614/\text{min}}$$

The cyclist's speed has decreased very significantly, but there is only a modest increase in the frequency of trucks passing him.

P17.52 We calculate the intensity of the speaker from

$$103 \text{ dB} = (10 \text{ dB}) \log\left(\frac{I}{10^{-12} \text{ W/m}^2}\right)$$

which gives $I = 2.00 \times 10^{-2} \text{ W/m}^2$

(a) We find the sound power output from

$$I = \frac{P}{4\pi r^2}$$

which gives

$$P = 4\pi r^2 I = 4\pi (1.60 \text{ m})^2 \left(2.00 \times 10^{-2} \text{ W/m}^2\right) = \boxed{0.642 \text{ W}}$$

(b) The efficiency of the speaker is

$$e = \frac{P_{out}}{P_{in}} = \frac{0.642 \text{ W}}{150 \text{ W}} = \boxed{0.004} \text{ or } \boxed{0.4\%}$$

P17.53 The flow of traffic at night is 1/20th that of the afternoon, so

$$P_2 = \frac{1}{20.0} P_1$$

The difference in sound level is

$$\beta_1 - \beta_2 = 10 \log\left(\frac{P_1}{P_2}\right)$$

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solving for the sound level at night gives

$$80.0 - \beta_2 = 10\log(20.0) = +13.0$$

$$\beta_2 = \boxed{67.0 \text{ dB}}$$

***P17.54** (a) We have $f' = \dfrac{fv}{v-u}$ and $f'' = \dfrac{fv}{v-(-u)}$. We then have

$$f' - f'' = fv\left(\frac{1}{v-u} - \frac{1}{v+u}\right)$$

$$\Delta f = \frac{fv(v+u-v+u)}{v^2-u^2} = \frac{2uvf}{v^2\left(1-\dfrac{u^2}{v^2}\right)} = \boxed{\frac{2\left(\dfrac{u}{v}\right)}{1-\dfrac{u^2}{v^2}} f}$$

(b) 130 km/h = 36.1 m/s

$$\Delta f = \frac{2(36.1 \text{ m/s})(400 \text{ Hz})}{(340 \text{ m/s})\left[1-\dfrac{(36.1 \text{ m/s})^2}{(340 \text{ m/s})^2}\right]} = \boxed{85.9 \text{ Hz}}$$

***P17.55** The sound speed is

$$v = 331 \text{ m/s} + (0.600 \text{ m/s} \cdot °\text{C})(26.0°\text{C}) = 347 \text{ m/s}$$

(a) Let t represent the time for the echo to return. Then

$$d = \frac{1}{2}vt = \frac{1}{2}(347 \text{ m/s})\left(24.0 \times 10^{-3} \text{ s}\right) = \boxed{4.16 \text{ m}}$$

(b) Let Δt represent the duration of the pulse:

$$\Delta t = \frac{10\lambda}{v} = \frac{10\lambda}{f\lambda} = \frac{10}{f} = \frac{10}{22.0 \times 10^6 \text{ s}^{-1}} = \boxed{0.455 \ \mu\text{s}}$$

(c) $$L = 10\lambda = \frac{10v}{f} = \frac{10(347 \text{ m/s})}{22.0 \times 10^6 \text{ s}^{-1}} = \boxed{0.158 \text{ mm}}$$

P17.56 (a) The sound "pressure" is extra tensile stress for one-half of each cycle. When it becomes (0.500%)(13.0 × 10^{10} Pa) = 6.50 × 10^8 Pa, the rod will break. Then, $\Delta P_{\max} = \rho v \omega s_{\max}$ and

$$s_{\max} = \frac{\Delta P_{\max}}{\rho v \omega} = \frac{6.50 \times 10^8 \text{ N/m}^2}{\left(8.92 \times 10^3 \text{ kg/m}^3\right)\left(5\,010 \text{ m/s}\right)\left(2\pi 500 \text{ s}^{-1}\right)}$$

$$= \boxed{4.63 \text{ mm}}$$

(b) From $s = s_{\max}\cos(kx - \omega t)$, differentiating gives

$$v = \frac{\partial s}{\partial t} = -\omega s_{\max}\sin(kx - \omega t)$$

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then

$$v_{\max} = \omega s_{\max} = \left(2\pi 500 \text{ s}^{-1}\right)(4.63 \text{ mm}) = \boxed{14.5 \text{ m/s}}$$

(c) $$I = \frac{1}{2}\rho v\left(\omega s_{\max}\right)^2 = \frac{1}{2}\rho v v_{\max}^2$$

$$= \frac{1}{2}\left(8.92\times 10^3 \text{ kg/m}^3\right)\left(5\,010 \text{ m/s}\right)\left(14.5 \text{ m/s}\right)^2$$

$$= \boxed{4.73\times 10^9 \text{ W/m}^2}$$

P17.57 The gliders stick together and move with final speed given by momentum conservation for the two-glider system:

$$m_1 v_1 + m_2 v_2 = m_1 v_1 + 0 = \left(m_1 + m_2\right)v$$

$$v = \frac{m_1 v_1}{m_1 + m_2} = \frac{(0.150 \text{ kg})(2.30 \text{ m/s})}{0.150 \text{ kg} + 0.200 \text{ kg}} = 0.986 \text{ m/s}$$

The missing mechanical energy is

$$\Delta K = \frac{1}{2}m_1 v_1^2 - \frac{1}{2}\left(m_1 + m_2\right)v^2$$

$$= \frac{1}{2}(0.150 \text{ kg})(2.30 \text{ m/s})^2 - \frac{1}{2}(0.350 \text{ kg})(0.986 \text{ m/s})^2$$

$$= 0.227 \text{ J}$$

We imagine one-half of 227 mJ going into internal energy and half into sound radiated isotropically in 7.00 ms. Its intensity 0.800 m away is

$$I = \frac{E}{At} = \frac{\frac{1}{2}(0.227 \text{ J})}{4\pi(0.800 \text{ m})^2\left(7.00\times 10^{-3} \text{ s}\right)} = 2.01 \text{ W/m}^2$$

Its intensity level is

$$\beta = (10 \text{ dB})\log\left(\frac{2.01 \text{ W/m}^2}{1.00\times 10^{-12} \text{ W/m}^2}\right) = 123 \text{ dB}$$

It is unreasonable, implying a sound level of 123 dB. Nearly all of the decrease in mechanical energy becomes internal energy in the latch.

P17.58 (a) The wave moves outward equally in all directions. (We can tell it is outward because of the negative sign in $1.36r - 2\,030t$.)

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(b) Its amplitude is inversely proportional to its distance from the center. Its intensity is proportional to the square of the amplitude, so the intensity follows the inverse-square law, with no absorption of energy by the medium.

(c) Its speed is constant at $v = f\lambda = \omega/k = (2030/\text{s})/(1.36/\text{m}) = 1.49$ km/s. By comparison to the table in the chapter, it can be moving through water at 25°C, and we assume that it is.

(d) Its frequency is constant at $(2030/\text{s})/2\pi = 323$ Hz.

(e) Its wavelength is constant at $2\pi/k = 2\pi/(1.36/\text{m}) = 4.62$ m.

(f) Its pressure amplitude is $(25.0\ \text{Pa}/r)$. Its intensity at this distance is

$$I = \frac{\Delta P_{\max}^2}{2\rho v} = \frac{\left[\left(25\ \text{N/m}^2\right)/r\right]^2}{2(1000\ \text{kg/m}^3)(1490\ \text{m/s})} = \frac{209\ \mu\text{W/m}^2}{r^2}$$

so the power of the source and the net power of the wave at all distances is

$$P = I4\pi r^2 = \left(\frac{2.09\times10^{-4}\ \text{W/m}^2}{r^2}\right)4\pi r^2 = 2.63\ \text{mW}$$

(g) Its intensity follows the inverse-square law; at $r = 1$ m, the intensity is $209\ \mu\text{W/m}^2$.

P17.59 (a) The speed of a compression wave in a bar is

$$v = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{20.0\times10^{10}\ \text{N/m}^2}{7\,860\ \text{kg/m}^3}} = \boxed{5.04\times10^{3}\ \text{m/s}}$$

(b) The signal to stop passes between layers of atoms as a sound wave, reaching the back end of the bar in time interval

$$\Delta t = \frac{L}{v} = \frac{0.800\ \text{m}}{5.04\times10^{3}\ \text{m/s}} = \boxed{1.59\times10^{-4}\ \text{s}}$$

(c) As described by Newton's first law, the rearmost layer of steel has continued to move forward with its original speed v_i for this time, compressing the bar by

$$\Delta L = v_i\Delta t = \left(12.0\ \text{m/s}\right)\left(1.59\times10^{-4}\ \text{s}\right) = 1.90\times10^{-3}\ \text{m}$$

$$= \boxed{1.90\ \text{mm}}$$

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(d) The strain in the rod is $\frac{\Delta L}{L} = \frac{1.90 \times 10^{-3} \text{ m}}{0.800 \text{ m}} = \boxed{2.38 \times 10^{-3}}$

(e) The stress in the rod is

$$\sigma = Y\left(\frac{\Delta L}{L}\right) = \left(20.0 \times 10^{10} \text{ N/m}^2\right)\left(2.38 \times 10^{-3}\right)$$

$$= \boxed{4.76 \times 10^{8} \text{ N/m}^2}$$

Since $\sigma > 400$ MPa , the rod will be permanently distorted.

(f) We go through the same steps as in parts (a) through (e), but use algebraic expressions rather than numbers:

The speed of sound in the rod is $v = \sqrt{\frac{Y}{\rho}}$

The back end of the rod continues to move forward at speed v_i for a time interval of $\Delta t = \frac{L}{v} = L\sqrt{\frac{\rho}{Y}}$, traveling distance $\Delta L = v_i \Delta t$ after the front end hits the wall.

The strain in the rod is $\frac{\Delta L}{L} = \frac{v_i t}{L} = v_i\sqrt{\frac{\rho}{Y}}$

The stress is then $\sigma = Y\left(\frac{\Delta L}{L}\right) = Y v_i\sqrt{\frac{\rho}{Y}} = v_i\sqrt{\rho Y}$

For this to be less than the yield stress, σ_y, it is necessary that the maximum speed be

$$\boxed{\frac{\sigma_y}{\sqrt{\rho Y}}}$$

P17.60 (a) Model your loud, sharp sound impulse as a single narrow peak in a graph of air pressure versus time. It is a noise with no frequency, wavelength, or period. It radiates away from you in all directions and some of it is incident on each one of the solid vertical risers of the bleachers.

The repeated reflections from the steps create a repetition frequency so that the ear/brain combination assigns a pitch to the sound heard by the listener.

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Suppose that, at the ambient temperature, sound moves at 343 m/s; and suppose that the horizontal width of each row of seats is 60 cm. Then there is a time delay of

$$\frac{0.60\text{ m}}{343\text{ m/s}} = 0.001\,7\text{ s}$$

between your sound impulse reaching each riser and the next. Whatever its material, each will reflect much of the sound that reaches it. The reflected wave sounds very different from the sharp pop you made.

(b) If there are twenty rows of seats, you hear from the bleachers a tone with twenty crests, each separated from the next in time by

$$\frac{2(0.60\text{ m})}{343\text{ m/s}} = 0.003\,5\text{ s}$$

This is the extra time for it to cross the width of one seat twice, once as an incident pulse and once again after its reflection. Thus, you hear a sound of definite pitch, with a period of about 0.0035 s, and frequency,

$$\frac{1}{0.003\,5\text{ s}} = 290\text{ Hz} \boxed{\sim \text{a few hundred Hz}}$$

(c) Wavelength

$$\lambda = \frac{v}{f} = \frac{343\text{ m/s}}{290\text{ s}^{-1}} = 1.2\text{ m} \sim \boxed{1\text{ m}}$$

(d) and duration

$$20(0.003\,5\text{ s}) \sim \boxed{0.1\text{ s}}$$

P17.61 Let f_e = 1 800 Hz represent the emitted frequency; v_e the speed of the skydiver; and f_g = 2 150 Hz the frequency of the wave crests reaching the ground.

(a) The skydiver source is moving toward the stationary ground, so we rearrange the equation $f_g = f_e\left(\frac{v}{v - v_e}\right)$ to give

$$v_e = v\left(1 - \frac{f_e}{f_g}\right) = (343\text{ m/s})\left(1 - \frac{1\,800\text{ Hz}}{2\,150\text{ Hz}}\right) = \boxed{55.8\text{ m/s}}$$

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(b) The ground now becomes a stationary source, reflecting crests with the 2 150-Hz frequency at which they reach the ground, and sending them to a moving observer, who receives them at the rate

$$f_{e2} = f_g\left(\frac{v+v_e}{v}\right) = (2\ 150\ \text{Hz})\left(\frac{343\ \text{m/s}+55.8\ \text{m/s}}{343\ \text{m/s}}\right)$$

$$= \boxed{2\ 500\ \text{Hz}}$$

P17.62 (a) The distance is larger by $240/60 = 4$ times. The intensity is 16 times smaller at the larger distance because the sound power is spread over a 42 times larger area.

(b) The amplitude is 4 times smaller at the larger distance because intensity is proportional to the square of amplitude.

(c) The extra distance is $(240-60)/45 = 4$ wavelengths. The phase is the same at both points because they are separated by an integer number of wavelengths.

P17.63 (a) If the velocity of the insect is v_x,

$$40.4\ \text{kHz} = (40.0\ \text{kHz})\frac{(343\ \text{m/s}+5.00\ \text{m/s})(343\ \text{m/s}-v_x)}{(343\ \text{m/s}-5.00\ \text{m/s})(343\ \text{m/s}+v_x)}$$

Solving, $v_x = \boxed{3.29\ \text{m/s}}$.

(b) Therefore, the bat is gaining on its prey at 1.71 m/s.

P17.64 When the observer is moving in front of and in the same direction as the source, $f' = f\dfrac{v-v_O}{v-v_S}$, where v_O and v_S are measured relative to the medium in which the sound is propagated. In this case the ocean current is opposite the direction of travel of the ships, and

$$v_O = 45.0\ \text{km/h} - (-10.0\ \text{km/h}) = 55.0\ \text{km/h} = 15.3\ \text{m/s}, \text{ and}$$

$$v_S = 64.0\ \text{km/h} - (-10.0\ \text{km/h}) = 74.0\ \text{km/h} = 20.55\ \text{m/s}$$

Therefore,

$$f' = (1\ 200.0\ \text{Hz})\frac{(1\ 533\ \text{m/s})-(15.3\ \text{m/s})}{(1\ 533\ \text{m/s})-(20.55\ \text{m/s})} = \boxed{1\ 204.2\ \text{Hz}}$$

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P17.65 (a) If the police car were at rest, the wavelength in air of its siren would be

$$\lambda = \frac{v}{f} = \frac{343\ \text{m/s}}{1\,000\ \text{s}^{-1}} = \boxed{0.343\ \text{m}}$$

(b) In front of the police car,

$$\lambda' = \frac{v}{f'} = \frac{v}{f}\left(\frac{v - v_S}{v}\right) = \frac{(343 - 40.0)\ \text{m/s}}{1\,000\ \text{s}^{-1}} = \boxed{0.303\ \text{m}}$$

(c) Behind the police car,

$$\lambda'' = \frac{v}{f''} = \frac{v}{f}\left(\frac{v + v_S}{v}\right) = \frac{(343 + 40.0)\ \text{m/s}}{1\,000\ \text{s}^{-1}} = \boxed{0.383\ \text{m}}$$

(d) The frequency heard by the speeder is

$$f' = f\left(\frac{v - v_O}{v - v_S}\right) = (1\,000\ \text{Hz})\frac{(343 - 30.0)\ \text{m/s}}{(343 - 40.0)\ \text{m/s}} = \boxed{1.03\ \text{kHz}}$$

P17.66 (a) The sound through the metal arrives first because it moves faster than sound in air.

(b) Each travel time is individually given by $t = L/v$. Then the delay between the pulses' arrivals is $\Delta t = L\left(\frac{1}{v_{\text{air}}} - \frac{1}{v_{\text{cu}}}\right) = L\frac{v_{\text{cu}} - v_{\text{air}}}{v_{\text{air}} v_{\text{cu}}}$ and the length of the bar is

$$L = \frac{v_{\text{air}} v_{\text{cu}}}{v_{\text{cu}} - v_{\text{air}}}\Delta t = \frac{(343\ \text{m/s})\left(3.56 \times 10^3\ \text{m/s}\right)}{(3\,560 - 343)\ \text{m/s}}\Delta t$$

$\boxed{L = 380\Delta t,\ \text{where } \Delta t \text{ is seconds and the length is in meters.}}$

(c) $L = (380\ \text{m/s})(0.127\ \text{s}) = \boxed{48.2\ \text{m}}$

(d) The answer becomes $L = \dfrac{\Delta t}{\frac{1}{343} - \frac{1}{v_r}}$, where v_r is the speed of sound in the rod in meters per second, Δt is in seconds, and L is in meters.

(e) As v_r goes to infinity, the travel time in the rod becomes negligible. The answer approaches $343\Delta t$, which is just the distance that the sound travels in air during the delay time.

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P17.67 (a) The Mach angle in the air is

$$\theta = \sin^{-1}\left(\frac{v_{\text{sound}}}{v_{\text{obj}}}\right) = \sin^{-1}\left(\frac{343}{20.0\times10^{3}}\right) = \boxed{0.983^\circ}$$

(b) At impact with the ocean,

$$\theta' = \sin^{-1}\left(\frac{1\,533}{20.0\times10^{3}}\right) = \boxed{4.40^\circ}$$

P17.68 The time interval required for a sound pulse to travel a distance L at a speed v is given by $t = \frac{L}{v} = \frac{L}{\sqrt{Y/\rho}}$. Using this expression, we find the travel time in each rod.

$$t_1 = L_1\sqrt{\frac{\rho_1}{Y_1}} = L_1\sqrt{\frac{2.70\times10^{3}\ \text{kg/m}^3}{7.00\times10^{10}\ \text{N/m}^2}} = L_1(1.96\times10^{-4}\ \text{s/m})$$

$$t_2 = (1.50 - L_1)\sqrt{\frac{11.3\times10^{3}\ \text{kg/m}^3}{1.60\times10^{10}\ \text{N/m}^2}}$$

$$= 1.26\times10^{-3}\ \text{s} - (8.40\times10^{-4}\ \text{s/m})L_1$$

$$t_3 = (1.50\ \text{m})\sqrt{\frac{8.80\times10^{3}\ \text{kg/m}^3}{11.0\times10^{10}\ \text{N/m}^2}} = 4.24\times10^{-4}\ \text{s}$$

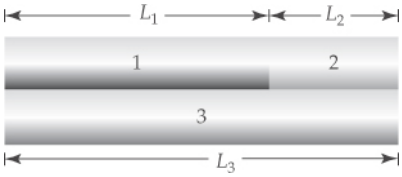
We require $t_1 + t_2 = t_3$, or

$$(1.96\times10^{-4}\ \text{s/m})L_1 + (1.26\times10^{-3}\ \text{s}) - (8.40\times10^{-4}\ \text{s/m})L_1 = 4.24\times10^{-4}\ \text{s}$$

This gives

$$L_1 = 1.30\ \text{m and } L_2 = (1.50\ \text{m}) - (1.30\ \text{m}) = 0.201\ \text{m}$$

The ratio of lengths is $\frac{L_1}{L_2} = \boxed{6.45}$.



ANS. FIG. P17.68

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P17.69 For the longitudinal wave $v_L = \left(\frac{Y}{\rho}\right)^{1/2}$

For the transverse wave $v_T = \left(\frac{T}{\mu}\right)^{1/2}$

If we require $\frac{v_L}{v_T} = 8.00$, we have $T = \frac{\mu Y}{64.0\rho}$ where $\mu = \frac{m}{L}$ and

$$\rho = \frac{\text{mass}}{\text{volume}} = \frac{m}{\pi r^2 L}$$

This gives

$$T = \frac{\pi r^2 Y}{64.0} = \frac{\pi\left(2.00\times10^{-3}\ \text{m}\right)^2\left(6.80\times10^{10}\ \text{N/m}^2\right)}{64.0}$$

$$= \boxed{1.34\times10^{4}\ \text{N}}$$

***P17.70** (a) Sound moves upwind with speed (343 – 15) m/s = 328 m/s. Crests pass a stationary upwind point at frequency 900 Hz. Then

$$\lambda = \frac{v}{f} = \frac{328\ \text{m/s}}{900\ \text{s}^{-1}} = \boxed{0.364\ \text{m}}$$

(b) By similar logic, $\lambda = \frac{v}{f} = \frac{(343+15)\ \text{m/s}}{900\ \text{s}^{-1}} = \boxed{0.398\ \text{m}}$

(c) The source is moving through the air at 15 m/s toward the observer. The observer is stationary relative to the air.

$$f' = f\left(\frac{v+v_o}{v-v_s}\right) = (900\ \text{Hz})\left(\frac{343\ \text{m/s}+0}{343\ \text{m/s}-15.0\ \text{m/s}}\right) = \boxed{941\ \text{Hz}}$$

(d) The source is moving through the air at 15 m/s away from the downwind firefighter. Her speed relative to the air is 30 m/s toward the source.

$$f' = f\left(\frac{v+v_o}{v-v_s}\right) = (900\ \text{Hz})\left(\frac{343\ \text{m/s}+30.0\ \text{m/s}}{343\ \text{m/s}-(-15.0\ \text{m/s})}\right)$$

$$= (900\ \text{Hz})\left(\frac{373\ \text{m/s}}{358\ \text{m/s}}\right) = \boxed{938\ \text{Hz}}$$

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Challenge Problems

P17.71 (a) If $v_O = 0$ m/s, then $f' = \dfrac{v}{v - v_S \cos\theta_S} f$.

Also, when the train is 40.0 m from the intersection, and the car is 30.0 m from the intersection,

$$\cos\theta_S = \frac{4}{5}$$

so $$f' = \frac{343 \text{ m/s}}{343 \text{ m/s} - 0.800(25.0 \text{ m/s})}(500 \text{ Hz}),$$

or $$f' = \boxed{531 \text{ Hz}}.$$

(b) Note that as the train approaches, passes, and departs from the intersection, θ_S varies from 0° to 180° and the frequency heard by the observer varies between the limits

$$f'_{\max} = \frac{v}{v - v_S \cos 0°} f = \frac{343 \text{ m/s}}{343 \text{ m/s} - 25.0 \text{ m/s}}(500 \text{ Hz})$$
$$= \boxed{539 \text{ Hz}}$$

to

$$f'_{\min} = \frac{v}{v - v_S \cos 180°} f = \frac{343 \text{ m/s}}{343 \text{ m/s} + 25.0 \text{ m/s}}(500 \text{ Hz})$$
$$= \boxed{466 \text{ Hz}}$$

(c) Now $v_O = +40.0$ m/s, and the train is 40.0 m from the intersection, and the car is 30.0 m from the intersection, so

$$\cos\theta_O = \frac{3}{5}$$

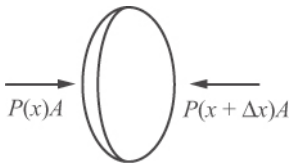
$$f' = \frac{343 \text{ m/s} + 0.600(40.0 \text{ m/s})}{343 \text{ m/s} - 0.800(25.0 \text{ m/s})}(500 \text{ Hz}) = \boxed{568 \text{ Hz}}$$

P17.72 (a) ANS. FIG. P17.72 shows a force diagram of an element of gas indicating the forces exerted on the left and right surfaces due to the pressure of the gas on either side of the element.

(b) Let $P(x)$ represent absolute pressure as a function of x. The net force to the right on the chunk of air is $+P(x)A - P(x + \Delta x)A$. Atmospheric pressure subtracts out, leaving

$$\left[-\Delta P(x + \Delta x) + \Delta P(x)\right]A = -\frac{\partial \Delta P}{\partial x}\Delta x A$$

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ANS. FIG. P17.72

The mass of the air is $\Delta m = \rho \Delta V = \rho A \Delta x$ and its acceleration is $\frac{\partial^2 s}{\partial t^2}$. So Newton's second law becomes

$$-\frac{\partial \Delta P}{\partial x}\Delta x A = \rho A \Delta x \frac{\partial^2 s}{\partial t^2}$$

(c) From the result above, we have

$$-\frac{\partial \Delta P}{\partial x}\Delta x A = \rho A \Delta x \frac{\partial^2 s}{\partial t^2} \quad \rightarrow \quad -\frac{\partial \Delta P}{\partial x} = \rho \frac{\partial^2 s}{\partial t^2}$$

Substituting $\Delta P = -(B \partial s / \partial x)$ (Eq. 17.3), we have

$$-\frac{\partial}{\partial x}\left(-B\frac{\partial s}{\partial x}\right) = \rho \frac{\partial^2 s}{\partial t^2} \quad \rightarrow \quad \frac{B}{\rho}\frac{\partial^2 s}{\partial x^2} = \frac{\partial^2 s}{\partial t^2}$$

(d) Into this wave equation we substitute a trial solution $s(x, t) = s_{\max} \cos(kx - \omega t)$. We find

$$\frac{\partial s}{\partial x} = -k s_{\max} \sin(kx - \omega t)$$
$$\frac{\partial^2 s}{\partial x^2} = -k^2 s_{\max} \cos(kx - \omega t)$$
$$\frac{\partial s}{\partial t} = +\omega s_{\max} \sin(kx - \omega t)$$
$$\frac{\partial^2 s}{\partial t^2} = -\omega^2 s_{\max} \cos(kx - \omega t)$$

$\frac{B}{\rho}\frac{\partial^2 s}{\partial x^2} = \frac{\partial^2 s}{\partial t^2}$ becomes

$$-\frac{B}{\rho}k^2 s_{\max} \cos(kx - \omega t) = -\omega^2 s_{\max} \cos(kx - \omega t)$$

This is true provided that $\frac{B}{\rho}k^2 = \omega^2 \rightarrow \frac{\omega}{k} = \sqrt{\frac{B}{\rho}}$, that is, provided it propagates with speed $v = \sqrt{\frac{B}{\rho}}$.

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P17.73 Figure 17.10 shows that each wavefront that passes the observer is spherical. Let T represent the period of the source vibration, and T_{MW} be the energy put into each wavefront during one vibration. Then $(Power)_{\text{avg}} = \dfrac{T_{\text{MW}}}{T}$. At the moment when the observer is at distance r in front of the source, he is receiving a spherical wavefront of radius $R_w = v\Delta t$, where Δt is the time interval since this energy was radiated. Since the wavefront was radiated, the source has moved forward distance $d_s = v_s \Delta t$, so the total distance the wavefront has traveled is

$$R_w = r + d_s \quad \rightarrow \quad v\Delta t = r + v_s \Delta t$$

therefore,

$$\Delta t = \frac{r}{v - v_S}$$

The surface area of the sphere is $4\pi R_w^2 = 4\pi (v\Delta t)^2 = \dfrac{4\pi v^2 r^2}{(v - v_S)^2}$. The energy per unit area emitted during one cycle and carried by one spherical wavefront is uniform with the value

$$I = \frac{T_{\text{MW}}}{A} = \frac{(Power)_{\text{avg}} T (v - v_S)^2}{4\pi v^2 r^2}$$

The energy carried by the wavefront passes the observer in the time interval $T' = 1/f'$, where f' is the Doppler-shifted frequency

$$f' = f\left(\frac{v}{v - v_S}\right) = \frac{v}{T(v - v_S)}$$

so the observer receives a wave with intensity

$$I = \left(\frac{T_{\text{MW}}}{A}\right)\frac{1}{T'} = \left(\frac{T_{\text{MW}}}{A}\right) f' = \left(\frac{(Power)_{\text{avg}} T (v - v_S)^2}{4\pi v^2 r^2}\right)\left(\frac{v}{T(v - v_S)}\right)$$

$$I = \boxed{\frac{(Power)_{\text{avg}}}{4\pi r^2}\left(\frac{v - v_S}{v}\right)}$$

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P17.2 (a) 1.27 Pa; (b) 170 Hz; (c) 2.00 m; (d) 340 m/**s**

P17.4 5.81 m

P17.6 1×10^{11} Pa

P17.8 (a) The speed gradually changes from $v = (331\text{ m/s})(1 + 27°\text{C}/273°\text{C})^{1/2} = 347\text{ m/s}$ to $(331\text{ m/s})\,(1 + 0/273°\text{C})^{1/2} = 331\text{ m/s}$, a 4.6% decrease. The cooler air at the same pressure is more dense; (b) The frequency is unchanged because every wave crest in the hot air becomes one crest without delay in the cold air; (c) The wavelength decreases by 4.6%, from $v/f = (347\text{ m/s})\,(4\,000/\text{s}) = 86.7$ mm to $(331\text{ m/s})(4\,000/\text{s}) = 82.8$ mm. The crests are more crowded together when they move more slowly.

P17.10 1.55×10^{-10} m

P17.12 (a) 153 m/s; (b) 614 m

P17.14 $d - \frac{1}{2}g\left(\sqrt{\frac{2(d-h)}{g}} - \Delta t - \frac{d-h}{v}\right)^2$ above the ground

P17.16 See P17.16 for complete solution.

P17.18 (a) 833 m; (b) 1.47 s

P17.20 (a) 5.00×10^{-5} W; (b) 3.00×10^{-3} J

P17.22 (a) $I_2 = \left(\frac{f'}{f}\right)^2 I_1$; (b) intensity is unchanged

P17.24 0.082 W/m^2

P17.26 150 dB

P17.28 (a) 332 J; (b) 46.4 dB

P17.30 $20\log\left(\frac{r_1}{r_2}\right)$

P17.32 (a) 65.0 dB; (b) 67.8 dB; (c) 69.6 dB

P17.34 (a) 1.76 kJ; (b) 108 dB

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P17.36 We assume that both lawn mowers are equally loud and approximately the same distance away. We found in Example 17.3 that a sound of twice the intensity results in an increase in sound level of 3 dB. We also see from the What If? section of that example that a doubling of loudness requires a 10-dB increase in sound level. Therefore, the sound of two lawn mowers will not be twice the loudness, but only a little louder than one!

P17.38 2.82×10^8 m/s

P17.40 (a) B; (b) positive; (c) negative; (d) 1 533 m/s; (e) 5.30×10^3 Hz

P17.42 (a) $\dfrac{vf}{v - A\sqrt{\dfrac{k}{m}}}$; (b) $\dfrac{vf}{v + A\sqrt{\dfrac{k}{m}}}$; (c) $\beta - (20\text{ dB})\log\left(1 + \dfrac{2A}{d}\right)$

P17.44 This is much faster than a human athlete can run.

P17.46 19.7 m

P17.48 5.67 mm

P17.50 (a) 0.232 m; (b) 8.41×10^{-8} m; (c) 13.8 mm

P17.52 0.642 W

P17.54 (a) $\dfrac{2\frac{u}{v}}{1 - \frac{u^2}{v^2}} f$; (b) 85.9 Hz

P17.56 (a) 4.63 mm; (b) 14.5 m/s; (c) 4.73×10^9 W/m^2

P17.58 (a) The wave moves outward equally in all directions; (b) Its amplitude is inversely proportional to its distance from the center. Its intensity is proportional to the square of the amplitude, so the intensity follows the inverse-square law, with no absorption of energy by the medium; (c) Its speed is constant $v = f\lambda = \omega/k = (2\,030/\text{s})(1.36/\text{m}) = 1.49\,\text{km/s}$. By comparison to the table, it can be moving through water at 25° C, and we assume it is; (d) Its frequency is constant at $(2\,030/\text{s})/2\pi = 323$ Hz; (e) Its wavelength is constant at $2\pi/\text{k} = 2\pi/(1.36/\text{m}) = 4.62$ m; (f) $P = I4\pi r^2 = \left(\dfrac{2.09 \times 10^{-4}\text{ W/m}^2}{r^2}\right)4\pi r^2 = 2.63$ mW; (g) Its intensity follows the inverse-square law; at r = 1 m, the intensity is $209\ \mu\text{W/m}^2$

P17.60 (a) The repeated reflections from the steps create a repetition frequency so that the ear/brain combination assigns a pitch to the sound heard by the listener; (b) ~ a few hundred Hz; (c) ~ 1 m; (d) ~ 0.1 s

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P17.62 (a) The distance is larger by 240/60 = 4 times. The intensity is 16 times smaller at the larger distance because the sound power is spread over a 42 times larger area; (b) The amplitude is 4 times smaller at the larger distance because intensity is proportional to the square of amplitude; (c) The extra distance is (240 – 60)/45 = 4 wavelengths. The phase is the same at both points because they are separated by an integer number of wavelengths

P17.64 1 204.2 Hz

P17.66 (a) The sound through the metal arrives first because it moves faster than sound in air; (b) $L = 380\Delta t,$ where Δt is in seconds and the length is in meters; (c) 48.2 m; (d) The answer becomes $L = \dfrac{\Delta t}{\dfrac{1}{343} - \dfrac{1}{v_r}}$ where v_r is the speed of sound in the rod in meters per second, Δt is in seconds, and L is in meters; (e) As v_r goes to infinity, the travel time in the rod becomes negligible. The answer approaches $343\Delta t$ which is just the distance that the sound travels in air during the delay time

P17.68 6.45

P17.70 (a) 0.364 m; (b) 0.398 m; (c) 941 Hz; (d) 938 Hz

P17.72 (a) See ANS. FIG P17.72; (b) See P17.72(b) for full explanation; (c) See P17.72(c) for full explanation; (d) See P17.72(d) for full explanation.

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